GTIIT Spring 2025 Calculus 1M
Workshop 1

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1 Inequalities and Absolute Values

1. Solve the inequality: $x^2 + x - 2 > 0$
2. Solve the inequality: $\dfrac{x+1}{x-3} \leq 0$
3. Solve the equation: $|2x + 5| = 9$
4. Solve the inequality: $|x - 4| < 3$
5. Solve the inequality: $|x^2 - 9| < 5$
6. Solve the compound inequality: $2 \leq |x + 1| < 5$

2 Graphs of Second-Degree Equations (Quadratic Functions)

(i) Sketch the graph of the circle with equation $x^2 + y^2 + x = 0$.

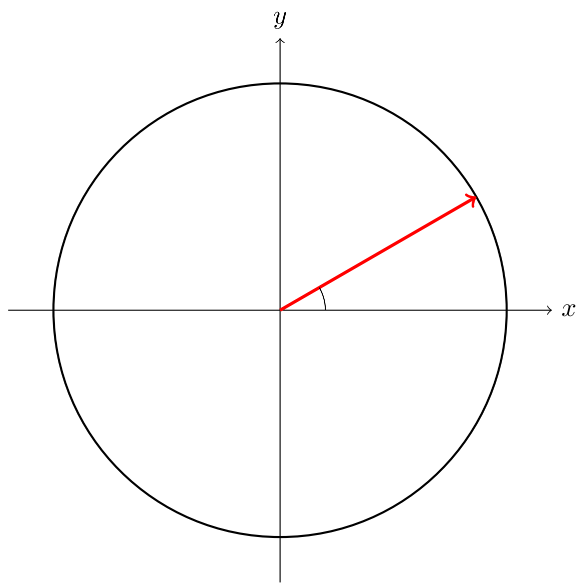
(ii) Under what conditions does the equation $x^2 + y^2 + ax + by + c = 0$ represent a circle?

(iii) Identify the type of curve represented by the equation $x^2 + y^2 + x = 0$ and sketch its graph.

(iv) Sketch the region bounded by the curves $y = 4 - x^2$ and $y = x^2 - 4$.

3 Trigonometric functions

Basic geometric interpretation: Draw and label in the following circle the geometric representation of the sine, cosine and tangent functions for an angle θ in the first quadrant.



(i) Solve the equation

$$\sin x = \cos x$$

for x in the interval $[0, 2\pi)$. *hint:* Divide both sides by $\cos x$ (assuming $\cos x \neq 0$).

(ii) Prove the identity

$$\tan x + \cot x = \frac{2}{\sin 2x}.$$

hint: Use the double-angle formula $\sin 2x = 2 \sin x \cos x$.

(iii) Solve the equation

$$\sin 2x = \sqrt{3} \cos x$$

for x in $[0, 2\pi)$. *hint:* Use the identity $\sin 2x = 2 \sin x \cos x$ and consider the cases $\cos x \neq 0$ and $\cos x = 0$.

(iv) Show that the expression $\sin x + \cos x$ can be rewritten as

$$\sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right).$$

hint: Recall the sine addition formula

$$\sin\left(x + \frac{\pi}{4}\right) = \sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4}.$$

(v) Prove the double-angle formula

$$\cos 2x = 2 \cos^2 x - 1.$$

hint: Use the standard double-angle formula $\cos 2x = \cos^2 x - \sin^2 x$.